Jacob Jameson

jacobjameson@g.harvard.edu
Cell: 475-777-9252
https://jacobjameson.com

HARVARD UNIVERSITY

Education

Harvard University
Ph.D. Health Policy (Decision Sciences Track), 2022–2027 (expected)

University of Chicago
M.S. Computational Analysis and Public Policy, 2020–2022

Johns Hopkins School of Education
Teaching Certification, Secondary Mathematics and Computer Science, 2018–2019

University of California, Santa Cruz
B.A. Economics and Mathematics, 2015–2018

Research Interests

Health care operations, causal inference, machine learning, decision science

References

Soroush Saghafian (Co-Chair)
Associate Professor of Public Policy
Harvard Kennedy School
soroush_saghafian@hks.harvard.edu

Sue Goldie (Co-Chair)
Roger Irving Lee Professor of Public Health
Harvard T.H. Chan School of Public Health
sue_goldie@harvard.edu

Robert S. Huckman
Albert J. Weatherhead III Professor of Business Administration
Harvard Business School
rhuckman@hbs.edu

Zachary Ward
Assistant Professor of Health Decision Science
Harvard T.H. Chan School of Public Health
zward@hsph.harvard.edu

Job Market Paper

The Impact of Batching Advanced Imaging Tests in Emergency Departments Preprint
Jameson, J. C., Saghafian, S., Huckman, R. S., Hodgson, N., & Baugh, J.
Invited 2nd round revision at Management Science

Using detailed electronic health record data from two major U.S. emergency departments (EDs), we examine practice variation across physicians to uncover the operational impact of discretionary batch ordering of imaging tests. We find that quasi-random assignment of a patient to an ED physician who is a “batcher” (top decile) versus a “sequencer” (bottom decile) causally increases the patient’s length of stay, time to disposition, and the number of imaging tests performed. Instrumental variable results show that discretionary batching increases length of stay by approximately 65%, increases imaging tests by 88%, and has no effect on 72-hour returns. Through a decomposition analysis, we find that approximately one-quarter of the LOS increase is attributable to increased imaging volume, with hospital admission as an additional contributing pathway, and the remainder consistent with increased coordination costs and decision complexity inherent to managing concurrent diagnostic workflows. Results suggest that discretionary batching may lead to clinical decision-making that introduces bottlenecks in patient flow. Conversely, standard practice, which preserves diagnostic flexibility by allowing information from initial tests to guide subsequent decisions, offers an “information gain” advantage over batch ordering: the information obtained from a prior test allows the elimination of the need to order some future tests. Put together, our findings indicate that discretionary batch ordering is not an efficient strategy for managing diagnostic imaging in emergency care, and interventions to reduce unnecessary batching may improve both operational performance and resource utilization.

Peer-Reviewed Publications

1. Transcranial Magnetic Stimulation Retreatment for Recurrent Depression: A Large Real-World Multisite Cohort Study.
   Acta Psychiatrica Scandinavica, 2026. PAPER
   Lyndon, S., Gibbons, J. B., Straub, L., Sung, M., **Jameson, J. C.**, Hathaway, D. B., Mulcahy, D. H., Elliott, C., Wood, R., Ruble, M., Szmulewicz, A. G., Yang, J., Aldis, R., Wyss, R., Wang, P. S., & Siddiqi, S. H.

2. Sex Differences in Life-Course Suicide Rates by State Firearm Policy Environment.
   American Journal of Preventive Medicine, 2025. PAPER
   Glasser, N. J., **Jameson, J. C.**, Abou Baker, N., Pollack, H. A., & Tung, E. L.

3. A National Study Among Diverse US Populations of Exposure to Prescription Medications with Evidence-Based Pharmacogenomic Information.
   Clinical Pharmacology & Therapeutics, 2025. PAPER
   Saulsberry, L., **Jameson, J. C.**, Gibbons, R. D., Dolan, M. E., Olopade, O. I., & O'Donnell, P. H.

4. Variation in Batch Ordering of Imaging Tests in the Emergency Department and the Impact on Care Delivery.
   Health Services Research, 2025. PAPER
   Jameson, J. C., Saghafian, S., Huckman, R. S., & Hodgson, N.

5. Genomic Heterogeneity and Ploidy Identify Patients with Intrinsic Resistance to PD-1 Blockade in Metastatic Melanoma.
   Science Advances, 2024. PAPER
   Tarantino, G., Ricker, C. A., Wang, A., Ge, W., Aprati, T. J., Huang, A. Y., Madha, S., Chen, J., Shi, Y., Glettig, M., Feng, C. H., Frederick, D. T., Freeman, S., Holovatska, M. M., Manos, M. P., Zimmer, L., Rösch, A., Zaremba, A., Livingstone, E., **Jameson, J. C.**, Saghafian, S., Lee, A., Zhao, K., Morris, L. G. T., Reardon, B., Park, J., Elmarakeby, H. A., Schilling, B., Giobbie-Hurder, A., Vokes, N. I., Buchbinder, E. I., Flaherty, K. T., Haq, R., Wu, C. J., Boland, G. M., Hodi, F. S., Van Allen, E. M., Schadendorf, D., & Liu, D.

6. Male Gender Expressivity and Diagnosis and Treatment of Cardiovascular Disease Risks in Men.
   JAMA Network Open, 2024. PAPER
   Glasser, N. J., **Jameson, J. C.**, Huang, E. S., Kronish, I. M., Lindau, S. T., Peek, M. E., Tung, E. L., & Pollack, H. A.

7. Associations of Adolescent School Social Networks, Gender Norms, and Adolescent-to-Young Adult Changes in Male Gender Expression with Young Adult Substance Use.
   Journal of Adolescent Health, 2024. PAPER
   Glasser, N. J., **Jameson, J. C.**, Tung, E. L., Lindau, S. T., & Pollack, H. A.

8. Does the Doctor-Patient Relationship Affect Enrollment in Clinical Research?
   Academic Medicine, 2023. PAPER
   Soo, J., **Jameson, J. C.**, Flores, A., Dubin, L., Perish, E., Afzal, A., Berry, G., DiMaggio, V., Krishnamoorthi, V. R., Porter, J., Tang, J., & Meltzer, D.

Working Papers

1. Reinforcement Learning for Repetitive Transcranial Magnetic Stimulation Protocol Selection in Treatment-Resistant Depression: A Multiclinic Cohort Study
   Invited revision at npj Digital Medicine
   Gibbons, J. B., Leenaerts, N., Siddiqi, S. H., Murphy, S., Straub, L., Hathaway, D. B., Yang, J., Szmulewicz, A. G., Sung, M., Aldis, R., **Jameson, J. C.**, Wyss, R., Wood, R., Ruble, M., Elliott, C., Mulcahy, D. H., Lyndon, S., & Wang, P. S.

2. Machine Learning to Personalize Psychotherapy Selection for Patients at Elevated Suicide Risk in a Large National Cohort
   Under review
   Jameson, J. C., Gibbons, J., Sung, M., Hathaway, D., Straub, L., Aldis, R., Lyndon, S., & Wang, P.

3. The Effect of Legalizing Online Sports Gambling on Population Mental Health.
Under review PREPRINT
Kavanagh, N. M., **Jameson, J. C.**, Pollack, H. A., & Glasser, N. J.

4. What Constitutes a Good Biomarker in Cancer Treatment: A Counterfactual Decision-Making Framework
In preparation
Jameson, J. C., Saghafian, S., Liu, D., Tarantino, G., & Gusev, S.

5. Allocating Scarce Clinician Outreach in Partner Notification
In preparation
Jameson, J. C., Saghafian, S., & Jónasson, J. O.

Research Funding

2024–present: National Research Service Award (NRSA) T32 Predoctoral Fellowship, National Institute of Mental Health (T32 MH125815), Brigham and Women's Hospital
Comparative Effectiveness Research in Suicide Prevention; Trainee; PIs: Miguel Hernán, Matthew Nock, Philip Wang

2026: Novartis Fund Project Award ($10,000), Harvard T.H. Chan School of Public Health
The Decision Science Learning Laboratory; Faculty Mentor: Sue J. Goldie
2026: Novartis Fund Conference Award ($1,000), Harvard T.H. Chan School of Public Health
2026: Graduate Student Council Conference Grant ($750), Harvard University

2024: Howard Raiffa Research Fund ($6,000), Center for Health Decision Science, Harvard T.H. Chan School of Public Health
AI-enabled software for causal reinforcement learning in melanoma treatment; Faculty Mentor: Soroush Saghafian

2022: Howard Raiffa Research Fund ($6,000), Center for Health Decision Science, Harvard T.H. Chan School of Public Health
Decision-analytic modeling of emergency department testing strategies; Faculty Mentor: Soroush Saghafian

Awards and Honors

2026: Distinction in Student Teaching Award, Harvard Kennedy School
2026: Dean's Award for Excellence in Student Teaching, Harvard Kennedy School
2026: Finalist, IOE-ISyE-MS&E Rising Stars Workshop, University of Michigan (People's Choice Award for Poster Presentation)
2026: Finalist, Harvard GSAS Three Minute Thesis (3MT®) Competition (1 of 10 finalists university-wide)

2025: Finalist, INFORMS Health Applications Society Best Student Paper Competition (1 of 4 finalists from 38 submissions)
2025: Distinction in Student Teaching Award, Harvard Kennedy School

2024: Dean's Award for Excellence in Student Teaching, Harvard Kennedy School
2024: Distinction in Student Teaching Award, Harvard Kennedy School
2024: Educational Innovation Scholar, Center for Health Decision Science, Harvard T.H. Chan School of Public Health

2023: Finalist, INFORMS Public Sector Operations Research Best Video Award

Presentations

Invited seminars: † Denotes scheduled
2026: Build Your Own RA (Research Agent), Harvard-wide Fellowship in Pediatric Health Services Research, Harvard Medical School†
2024: Causal Inference and Suicide Prevention, Center for Suicide Research and Prevention, Mass General Brigham and Harvard University

Conference presentations:
2026: INFORMS Annual Meeting (San Francisco, CA; invited)[†]; INFORMS Healthcare Conference (Raleigh, NC; invited); POMS Annual Conference (Reno, NV; contributed); IOE-ISyE-MS&E Rising Stars Workshop, University of Michigan (contributed); Harvard GSAS Three Minute Thesis Finals
2025: INFORMS Annual Meeting (Atlanta, GA; two invited talks)
2024: Society for Medical Decision Making Annual Meeting (Boston, MA; contributed); INFORMS Annual Meeting (Seattle, WA; invited)
2023: INFORMS Annual Meeting (Phoenix, AZ; invited); INFORMS Healthcare Conference (Toronto, ON; invited)

Teaching

Economics for Public Policy, Harvard Kennedy School
Instructor of Record, PPIA Junior Summer Institute, 2023–2026 (3x)

Math Camp for Incoming PhD Students, Harvard Kennedy School
Instructor of Record, 2023–2026 (4x)

Decision Science for Public Health, Harvard T.H. Chan School of Public Health
Teaching Fellow, 2023–2026 (4x)

Big Data and Machine Learning, Harvard Kennedy School
Teaching Fellow, 2023–2025 (4x)

Game Theory, Harvard Kennedy School
Teaching Fellow, 2023, 2024, 2026 (3x)

Resources, Incentives, and Choices I: Markets and Market Failures, Harvard Kennedy School
Teaching Fellow, 2022–2025 (4x)

Data and Programming for Policymakers, Harvard Kennedy School
Teaching Fellow, 2023 (1x)

Data, Quantitative Methods, and Applications in Health Services Research, Center for Translational Science, University of Chicago
Co-Instructor, 2021–2023 (3x)

Coding Lab for Public Policy, Harris School of Public Policy, University of Chicago
Instructor of Record, 2021 (1x)

Machine Learning for Public Policy, Department of Computer Science, University of Chicago
Teaching Assistant, 2022 (1x)

Mathematics for Data Analysis and Computer Science, Department of Computer Science, University of Chicago
Teaching Assistant, 2022 (1x)

Introductory Finance, Booth School of Business, University of Chicago
Teaching Assistant, 2020–2022 (3x)

Data Analysis in R and Python, Booth School of Business, University of Chicago
Teaching Assistant, 2022 (1x)

Professional Experience

VA Greater Los Angeles Healthcare System, Department of Veterans Affairs
Research Statistician (Without Compensation appointment), 2026–present

Aerospace Technical Services
Health Policy Consultant, 2025–present

Division of Pharmacoepidemiology and Pharmacoeconomics, Department of Medicine, Brigham and Women's Hospital & Harvard Medical School
Research Trainee, 2024–present

Center for Health and the Social Sciences, University of Chicago
Data Scientist, 2020–2022

Teach For America, Achievement First Amistad Academy
7th/8th Grade Mathematics Teacher, 2018–2020

Academic Service

Leadership:
2026: Teaching Statistics in the Age of Agentic AI, Harvard Kennedy School Workshop, Organizing Team
2026: Short Course Co-Chair, SMDM Annual Meeting
2022–2024: Health Applications Society Student Liaison, INFORMS

Session chair:
2025: INFORMS Annual Meeting (Atlanta, GA)
2024: INFORMS Annual Meeting (Seattle, WA; two sessions)
2023: INFORMS Annual Meeting (Phoenix, AZ); INFORMS Healthcare Conference (Toronto, ON)

Short courses and workshops:
2026: Build Your Own RA (Research Agent): Introduction to Agentic Coding and Custom Research Workflows, SMDM (Instructor)†
2024: An Introduction to Reproducible Programming and Project Management, SMDM Annual Meeting (Instructor)
2023: Transparent Programming: Important Habits for Reproducibility and Research Integrity, SMDM Annual Meeting (Instructor)

Ad hoc reviewer: Medical Decision Making, Journal of Public Health and Emergency

Conference referee: Society for Medical Decision Making Annual Meeting; International Conference on Computational Social Science (IC^2S^2)

Professional Memberships

Institute for Operations Research and the Management Sciences (INFORMS); Society for Medical Decision Making (SMDM)

Software Skills

R, Python, Stata, SQL; D3, Git, Docker, TreeAge